

Computer Science & IT

Database Management System



Query Languages

Lecture No. 12



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Recap of Previous Lecture

- ✓ **Topic** Introduction to relational calculus
- ✓ **Topic** Tuple relational calculus
- ✓ **Topic** Practice questions



Topics to be Covered



✓
Topic

Practice questions

✓
Topic

Domain relational calculus (DRC)



Topic : Unsafe TRC query



$\{ t \mid t \notin \text{Relation} \}$

or

$\{ t \mid \neg(t \in \text{Relation}) \}$

Here we are trying to select all the tuples which do not belong to the Relation

A Complement of
"a set of tuples
belonging to relation."

If domain of an attribute happens to be infinite then such query may produce infinite number of tuples in output.

- Unsafe TRC query may produce infinite number of tuples.
- We can never write an Equivalent relational algebra query with respect to an unsafe TRC query.
- For Every safe TRC query we can always write an Equivalent relational algebra query.

Power of TRC query $\neq$ Power of Relational Algebra
(Safe + Unsafe)

Power of Safe TRC query $\equiv$ Power of Relational Algebra

#Q. Which of the following relational calculus expressions is not safe?

- A. $\{t | \exists u \in R1 (t[A] = u[A]) \wedge \neg \exists s \in R2 (t[A] = s[A])\}$
 $\uparrow$ $\forall s \in R2 (t[A] \neq s[A])$
- B. $\{t | \forall u \in R1 (u[A] = "x" \Rightarrow \exists s \in R2 (t[A] = s[A] \wedge s[A] = u[A]))\}$
- ☒ C. $\{t | \neg(t \in R1)\}$
- D. $\{t | \exists u \in R1 (t[A] = u[A]) \wedge \exists s \in R2 (t[A] = s[A])\}$

#Q.

Consider the relation employee (name, sex, supervisorName) with name as the key. Supervisor Name gives the name of the supervisor of the employee under consideration. What does the following Tuple Relational Calculus query produce?

"A"
"B"
 $\{e.name \mid \text{employee}(e) \wedge (\forall x) [\neg \text{employee}(x) \vee x.supervisorName \neq e.name \vee x.sex = \text{"male"}]\}$

Permanently false

A. Names of employees with a male supervisor.

B. Names of employees with no immediate male subordinates

C. Names of employees with no immediate female subordinates

D. Names of employees with a female supervisor.

name	Sex	Sup.
A	M	B
B	F	A
C	F	D
D	F	NULL
E	F	NULL

#Q. Consider the relation employee (name, sex, supervisorName) with name as the key. Supervisor Name gives the name of the supervisor of the employee under consideration. What does the following Tuple Relational Calculus query produce?

{e.name | employee(e)

$\wedge [\forall x [\neg \text{employee}(x) \vee x.\text{supervisorName} \neq e.\text{name} \vee x.\text{sex} = \text{"male"}]]$

$\sim \exists x [\text{employee}(x) \wedge x.\text{supervisorName} = e.\text{name} \wedge x.\text{sex} = \text{female}]$

A. Names of employees with a male supervisor.

B. Names of employees with no immediate male subordinates

✓ C. Names of employees with no immediate female subordinates

D. Names of employees with a female supervisor.

#Q. Which of the following tuple relational calculus expression(s) is/are equivalent to $\forall t \in r(P(t))$?

I $\neg \exists t \in r(P(t))$

II $\exists t \notin r(P(t))$

III $\neg \exists t \in r(\neg P(t))$

IV $\neg \exists t \notin r(\neg P(t))$

A. I only

B. II only

C. III only

D. III and IV only

for all tuples t belonging to R
Predicate $P(t)$ must be true

III
i.e., not for atleast one tuple t of R ,
predicate $P(t)$ should be false

$$\begin{aligned} & \forall t \in R (P(t)) \\ & \sim \sim [\forall t \in R (P(t))] \\ & \sim \exists t \in R (\sim P(t)) \end{aligned}$$

#Q.

Consider The Following Relational Scheme

Student (school-id, sch-roll-no, sname, saddress)

School (school-id, sch-name, sch-address, sch-phone)

Enrolment (school-id, sch-roll-no, erollno, examname)

ExamResult (erollno, examname, marks)

Consider the following tuple relational calculus query

$\{t \mid \exists E \in \text{Enrolment } t = E.\text{school-id} \wedge \dots\}$ *will be true*

$\{x \mid x \in \text{Enrolment} \wedge x.\text{school-id} = t \wedge (\exists B \in \text{ExamResult } B.\text{erollno} = x.\text{erollno} \wedge$

$B.\text{examname} = x.\text{examname} \wedge B.\text{marks} > 35) \mid \div$

$\{x \mid x \in \text{Enrolment} \wedge x.\text{school-id} = t\} \mid * 100 > 35$

If a student needs to score more than 35 marks to pass an exam what does the query return?

True
then

if it is
false then
predicate is false

#Q.



✗ A. The empty set

✗ B. schools with more than 35% of its students enrolled in some exam or the other

✓ C. schools with a pass percentage above 35% over all exams taken together

D. schools with a pass percentage above 35% over each exam

#Q.

✓ Consider a database that has the relation schemas
EMP(EmpId, EmpName, DeptId) and

✓ DEPT(DeptName, DeptId).

Note that the DeptId can be permitted to a NULL in the relation EMP. Consider the following queries on the database expressed in tuple relational calculus.

I. $\{t \mid \exists u \in \text{EMP} (t[\text{EMPName}] = u[\text{EMPName}] \wedge \forall v \in \text{DEPT} (t[\text{DeptId}] \neq \text{DeptId}))\}$

II. $\{t \mid \exists u \in \text{EMP} (t[\text{EMPName}] = u[\text{EMPName}] \wedge \exists v \in \text{DEPT} (t[\text{DeptId}] \neq \text{DeptId}))\}$

III. $\{t \mid \exists u \in \text{EMP} (t[\text{EMPName}] = u[\text{EMPName}] \wedge \exists v \in \text{DEPT} (t[\text{DeptId}] = \text{DeptId}))\}$

Which of the above queries are safe?

- A. I and II only
- B. I and III only
- C. II and III only
- ✓ D. I, II and III



Topic : Syntax of DRC



- Each query is an expression of the form
$$\{ \langle \underline{x_1, x_2, \dots, x_n} \rangle \mid \underline{P(x_1, x_2, \dots, x_n)} \}$$
- Where $x_1, x_2, \dots, x_n$ represent output domain variables.
- $P(x_1, x_2, \dots, x_n)$ is the predicate formula.

eg. Consider the relation
Emp (Eid, name, Salary)
retrieve all the tuples of Emp table

• TRC: $\{ t \mid t \in \text{Emp} \}$
w.r.t. (row)

• DRC $\{ \langle \underbrace{E.N.S} \rangle \mid \langle E.N.S \rangle \in \text{Emp} \}$
We need all three Column in o/p
E is 1st Column of Emp
N is 2nd Column of Emp
S is 3rd Column of Employee

eg. Consider the relation
Emp (Eid, name, Salary)
retrieve Eids from Emp table

• TRC: $\{ t.Eid \mid t \in \text{Emp} \}$

• DRC $\{ \langle E \rangle \mid \langle E, N, S \rangle \in \text{Emp} \}$

E is 1st Column of Emp
N is 2nd Column of Emp
S is 3rd Column of Employee

eg. Consider the relation
Emp (Eid, name, Salary)
retrieve distinct names from Emp table

• TRC: $\{ t.name \mid t \in Emp \}$

• DRC $\{ \langle N \rangle \mid \langle E, N, S \rangle \in Emp \}$

E is 1st Column of Emp
N is 2nd Column of Emp
S is 3rd Column of Employee

#Q. Consider the following relational schemas
Employee(Empld, EmpName, salary)

Select distinct EmpName from Employee table where salary > 2000

TRC: $\{ t.\text{EmpName} \mid t \in \text{Employee} \wedge t.\text{Salary} > 2000 \}$

DRC: $\{ \langle N \rangle \mid \langle E, N, S \rangle \in \text{Employee} \wedge \langle S \rangle > 2000 \}$

#Q. Consider the following relational schemas
Student(Rollno, name, marks)

Select distinct name and marks of the student from Student table where
Rollno = 10

TRC: $\{ t.name, t.marks \mid t \in \text{Student} \wedge t.Rollno = 10 \}$

DRC: $\{ \langle N, M \rangle \mid \langle R, N, M \rangle \in \text{Student} \wedge \langle R \rangle = 10 \}$

$\{ \langle N, M \rangle \mid \langle 10, N, M \rangle \in \text{Student} \}$

#Q. Consider the following relational schema.

Students(rollno: integer, sname: string)

Courses(courseno: integer, cname: string)

Registration(rollno: integer, courseno: integer, percent: real)

Which of the following queries are equivalent to this query in English?

“Find the distinct names of all students who score more than 90% in the course numbered 107”

- I. SELECT DISTINCT S.sname FROM Students as S, Registration as R WHERE R.rollno=S.rollno AND R.courseno=107 AND R.percent >90
- II. $\Pi_{\text{sname}}(\sigma_{\text{courseno}=107 \wedge \text{percent} > 90}(\text{Registration} \bowtie \text{Students}))$
- III. $\{ T \mid \exists S \in \text{Students}, \exists R \in \text{Registration} (S.\text{rollno} = R.\text{rollno} \wedge R.\text{courseno} = 107 \wedge R.\text{percent} > 90 \wedge T.\text{sname} = S.\text{sname}) \}$
- IV. $\{ \langle S_N \rangle \mid \exists S_R, \exists R_P (\langle S_R, S_N \rangle \in \text{Students} \wedge \langle S_R, 107, R_P \rangle \in \text{Registration} \wedge R_P > 90) \}$

- A. I, II, III and IV
- B. I, II and III only
- C. I, II and IV only
- D. II, III and IV only

$\{ \langle S_N \rangle \mid \langle S_R, S_N \rangle \in \text{Student} \wedge \langle R_R, R_C, R_P \rangle \in \text{Reg.} \wedge \langle S_R \rangle = \langle R_R \rangle \wedge \langle R_C = 107 \rangle \wedge \langle R_P \rangle > 90 \}$

107



2 mins Summary



✓
Topic

PYQ of TRC

✓
Topic

DRC

✓
Topic

PYQ of DRC



THANK - YOU